

How well does AMANA AI find our own knowledge?

Testing and improving the knowledge search that AMANA AI uses to answer questions · 30 September 2026

The bottom line. When we started measuring, the knowledge search was returning **no results at all** for essentially every realistic question a colleague would ask. It looked like it was working, but it was not.

It now finds the correct source among the first ten results **92% of the time**, and puts it in first place more than half the time. For our reports and studies specifically, it finds the right document **100% of the time**. Questions asked in Indonesian now find English documents and vice versa, which previously failed completely.

Three things still need attention, listed near the end. The most important is that the system cannot yet tell the difference between "I found the answer" and "I found something vaguely related" — so it never says "I don't know".

What we actually tested

We built a set of **100 test questions** drawn directly from AMANA's own material — our reports, studies, staff profiles, past projects and partner contacts. For each question we recorded which document or record genuinely contains the answer, so we can check whether the search finds it.

The questions were written to resemble what colleagues actually type, not keyword searches. They were deliberately reworded so they do not copy phrasing from the source — otherwise we would only be testing word-matching, not whether the system understands the question.

What the set contains	Questions	Why it is there
Questions about our reports and studies	70	Spread so every one of our 35 documents is covered, not just the largest
Questions about staff, partners and past projects	15	Search covers these too, and they behave very differently
Questions our knowledge <i>cannot</i> answer	15	To check the system admits it has nothing, instead of guessing

Half the questions are in English and half in Indonesian, balanced across every category. Twenty of them deliberately ask in one language about a document written in the other — a realistic situation for us, and one the old search handled very badly.

The results

Three points in time: where we started, after a first repair, and where it stands today. Every figure is measured against the same 100 questions, so the comparison is fair. Each measure is explained in the next section.

What we measure	At the start	After first fix	Today
Searches that returned nothing at all	100%	0%	0%
Correct source shown first	0%	25%	54%
Correct source in the top three	0%	52%	75%
Correct source in the top ten	0%	71%	92%
Typical position score (0 to 1)	0.00	0.40	0.67
Questions crossing between Indonesian and English	0%	40%	100%

What each measure means

Searches that returned nothing at all (technical name: zero-hit rate)

Out of every 100 searches, how many came back completely empty?

This is a health check rather than a quality measure — it does not ask whether the answer was *right*, only whether anything came back. At the start this was **100%**: the search demanded that every single word of a question appear in one document, which almost never happens in a real sentence. It is now **0%**. Anything above zero here means something is broken, not merely imperfect.

Correct source shown first (recall@1)

How often is the very first result the right one?

This matters because AMANA AI pays most attention to the first result, and sometimes opens only that one. Now **54%** — up from nothing. This is the measure with the most room left to improve: the right document is usually retrieved, but not always placed at the top.

Correct source in the top three (recall@3)

How often does the right answer appear among the first three results?

A realistic picture of what AMANA AI reads carefully before answering. Now **75%**.

Correct source in the top ten (recall@10)

How often does the right answer appear anywhere in the returned results?

This is the single most important figure in the report. If the right document is never retrieved, AMANA AI cannot possibly base its answer on it — no amount of clever wording or model improvement can recover information that was never found. This number is therefore a hard ceiling on how reliable any answer can be.

Now **92%** . In plain terms: for roughly 8 questions in 100 the right source is still missed, and for those the answer cannot be trusted regardless of how confident it sounds.

Typical position score (MRR, or Mean Reciprocal Rank)

A single number between 0 and 1 summarising how near the top the right answer usually sits.

It works by scoring each question by the position of the correct result — first place scores 1.00, second place 0.50, third 0.33, tenth 0.10, and not found at all scores 0 — then averaging across all questions. A score of 1.00 would mean the right answer is always first; 0 would mean it is never found.

Now **0.67** , from 0.00. This is the best single figure to track month to month, because it moves both when the system finds more answers *and* when it ranks them better. The other measures can each stay flat while quality quietly drifts.

The biggest single improvement: our two languages now work together

AMANA's knowledge is a mix of Indonesian and English documents, and colleagues ask questions in both languages. Previously the search compared questions to documents purely by matching words. An Indonesian question and an English report share almost no words, so those searches failed almost entirely — they succeeded 0% of the time at the start, and still only 40% after the first repair.

The search now also compares questions and documents **by meaning**, not just by wording, using a method that works across both languages. Cross-language questions now succeed **100% of the time** in our test set.

In practice this means a colleague can ask in Bahasa Indonesia and still be pointed to the right English report — and the reverse. Given how our material is split, this was a substantial blind spot.

How different kinds of information perform

Search covers five kinds of record, and they do not perform equally. This is worth knowing, because it tells us where to trust the system today and where to double-check it.

Kind of record	In collection	Found in top ten	Assessment
Reports, studies and other documents	35	100%	Reliable. This is the bulk of our knowledge.
Past projects	10	100%	Reliable.

Kind of record	In collection	Found in top ten	Assessment
Internal knowledge notes	8	100%	Reliable.
Staff profiles and expertise	104	50%	Improves to 100% when the question is aimed specifically at people — see below.
Partner and contact organisations	220	0%	Does not work. This is a data problem, not a search problem.

On staff searches. When a question is explicitly aimed at people, the right colleague is found 100% of the time. The 50% figure applies when the question is open-ended, because documents are long and rich in words while a staff profile is short, so documents tend to fill all the available result slots. We have made two changes for this: reserving some result slots for people and organisations, and instructing AMANA AI to narrow its search when a question is clearly about staff. We expect this to improve further in real use.

What still needs attention

1. The system does not know when it does not know

We tested 15 questions our knowledge genuinely cannot answer — about sectors and topics AMANA has no material on. **All 15 returned confident-looking results anyway.** Nothing in the results distinguishes "this is the answer" from "this is the closest thing I could find, and it is not relevant".

This is the most important remaining risk, because it is how a plausible but wrong answer reaches a colleague. A proposal could be written on a claim the knowledge base never actually supported.

The fix, which is next on our list: have the search report how strong its evidence is — strong, moderate or weak — so AMANA AI can say "I could not find anything on this" or answer with appropriate caution instead of stating it as fact. Our measurements show the underlying signal to do this already exists; it simply is not being passed on yet.

2. Partner and contact records are too thin to search

With 220 partner organisations, searches for them fail. The cause is not the search: most records contain little more than an organisation name, an affiliation and a category. There is not enough written down to tell one from another.

No amount of technical work fixes this. It needs a short description of what each organisation actually does, added through the dashboard. Even one or two sentences each would make them findable.

3. Document details are unreliable, so sources may look wrong

Separately from search quality, we found that the information recorded *about* our documents has errors. Some examples:

- Several documents are dated in the future — one is recorded as 2045.
- One document's title is not a title at all, but a line copied from a table of contents.
- The same client appears under two different names ("Kemenkes" and "Kementerian Kesehatan"), so filtering by client splits them.
- Section headings quoted alongside answers are often fragments rather than real headings.
- About a third of our documents have no practice group recorded.

The practical effect: AMANA AI may find exactly the right passage but present it with a wrong-looking date, title or heading — which makes correct answers look untrustworthy. These originate in the automatic document intake process, and most cannot currently be corrected by hand because the dashboard has no screen for editing these records.

What happens next

1. **Teach the system to say "I don't know."** Report evidence strength with every search so weak matches are flagged rather than presented as fact. Highest priority.
2. **Enrich the partner records** with a sentence or two describing each organisation.
3. **Correct the document details** — dates, titles, client names — and add a dashboard screen so they can be maintained going forward.
4. **Improve first-place accuracy** from 54%. The right document is nearly always found; the remaining work is ordering results better.
5. **Begin measuring the answers themselves**, not just the search. Now that the right sources are reliably retrieved, we can start scoring whether AMANA AI's written answers are properly supported by them, and build the quality dashboard on top of that.

A note on how this was measured

Everything in this report is measured, not estimated. The same 100 questions are used every time, the test is repeatable on demand, and it can be re-run in under a minute without cost. That means any future change to the search can be checked against these figures before it reaches colleagues, rather than being judged by impression.

Two honest limitations. First, 100 questions is a solid sample but not a large one — the figures for staff, partners and past projects rest on only a few questions each and should be treated as indicative. Second, these questions were generated from our own documents, which means they test whether the system can find material it holds; they cannot tell us what colleagues will ask that we have no material for at all.